Alpha Diversity Analysis

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Contents

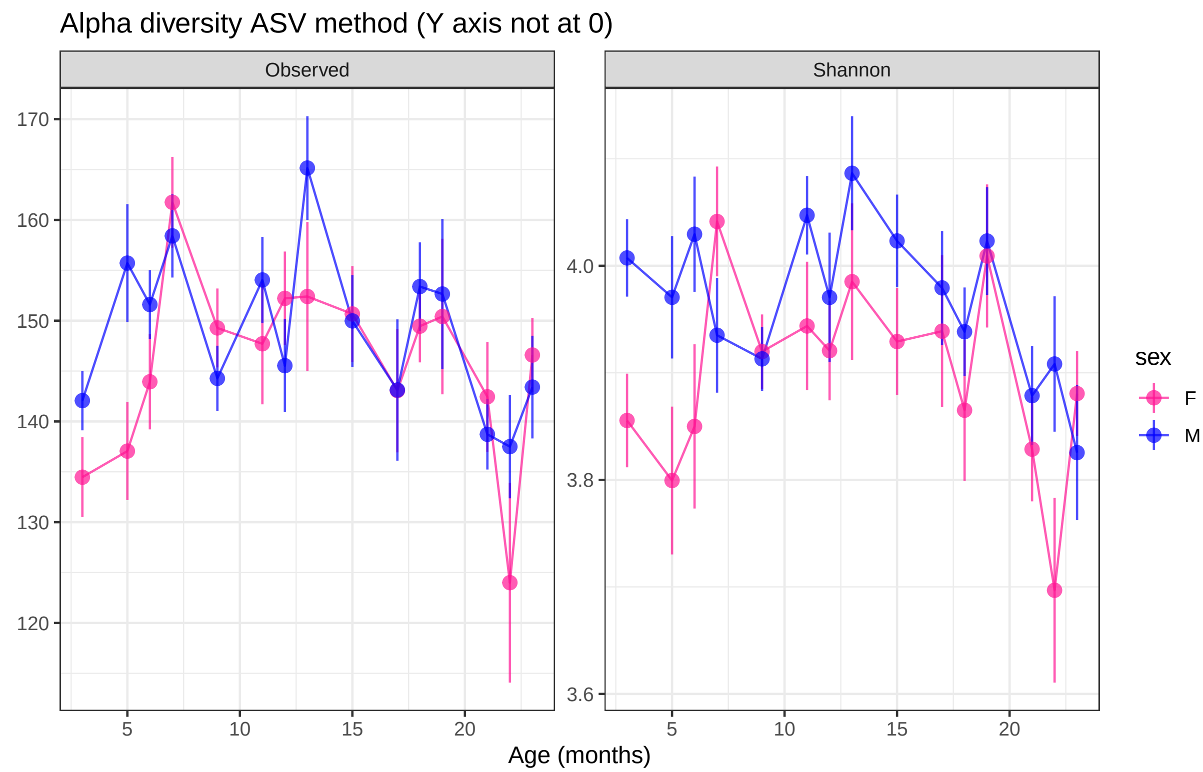
Based on the below results, I determined that sex difference in alpha diversity were minimal. The alpha diversity and age do not have a linear relationship.

```r
## alpha diversity ====
library(lme4)
library(lmerTest)
library(knitr)
library(dplyr)
library(phyloseq)
library(vegan)
library(emmeans)
library(tibble)
library(reshape2)
library(tidyverse)
library(qiime2R)
library(ggpubr)
library(ggplot2)

theme_set(theme_bw())

load("qiime_f.RData")
metadata_df<-as.data.frame(sample_data(qiime_f))
metadata_df$SampleID<-rownames(metadata_df)
metadata_df<-as.matrix(metadata_df)

qiime_f_rare<-rarefy_even_depth(qiime_f,sample.size = min(sample_sums(qiime_f)))

alpha.div.df <- estimate_richness(
  qiime_f_rare,
  measures=c("Shannon", "Observed")) %>%
  rownames_to_column("SampleID") %>%
```

```r
  merge(metadata_df, by="SampleID")

alpha.div.long.df <- alpha.div.df %>%
  pivot_longer(c(Shannon, Observed), names_to="metric")

alpha.div.long.df$month_n<-as.numeric(alpha.div.long.df$month_n)

# summarize across sexes
alpha.div.plot.df <- alpha.div.long.df %>%
  dplyr::filter(metric %in% c("Shannon", "Observed")) %>%
  dplyr::filter(month_n <= 40) %>% # omit the few 46 month samples

  # compute mean and SD across diets (separately for each metric)
  group_by(metric, sex, month_n) %>%
  summarise(n=n(),
            mean = mean(value),
            sd = sd(value),
            sem = sd(value)/sqrt(n()), .groups="drop") %>%

  # make sure we have at least 3 observations at this age combination
  dplyr::filter(n >= 2) #%>%

#plot Fig 2 alpha ====
alpha.div.plot.df <- alpha.div.plot.df %>%
  arrange(metric, sex, month_n)
```

Alpha diversity plot

```r
alpha.div.plot.df %>%
  ggplot(aes(
    x = month_n, y = mean,
    ymin = mean - sem, ymax = mean + sem,
    color = sex
  )) +
  geom_path(alpha = 0.7) +
  geom_pointrange(alpha = 0.7) +
  facet_wrap(~ metric, scales = "free", nrow = 1) +
  scale_color_manual(values = c("deeppink", "blue")) +
  labs(
    x = "Age (months)",
    y = "",
    title = "Alpha diversity ASV method (Y axis not at 0)"
  ) +
  theme_bw()
```



LMER for alpha

```
df_wide <- alpha.div.long.df %>%
  pivot_wider(names_from = metric, values_from = value)
```

Q Are there differences between sex and month, and if so, when?

From model1, we see that there is no interaction of sex and month, i.e. temporal changes in Shannon diversity were similar in males and females. However, the model shows that there are sex differences and there are differences with age of mice. M13 and M22 were significantly different from baseline M03.

Because there was no significant sex × month interaction, we next examined which months differed from the baseline month (M03) within each sex using estimated marginal means (emmeans).

```
model1_interaction <- lmer(Shannon ~ sex * month + (1 | mousegroup.x) + (1 | mouse), data = df_wide)
drop1(model1_interaction,test="Chisq")
```

```
## Single term deletions using Satterthwaite's method:
##
## Model:
## Shannon ~ sex * month + (1 | mousegroup.x) + (1 | mouse)
##             Sum Sq  Mean Sq NumDF  DenDF F value  Pr(>F)
## sex:month 1.1581 0.082721    14 560.53  1.5337 0.09429 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA for model1 (Shannon with all cohorts)

```r
model1 <- lmer(Shannon ~ sex + month + (1 | mousegroup.x) + (1 | mouse), data = df_wide)
drop1(model1,test="Chisq")
```

```
## Single term deletions using Satterthwaite's method:
##
## Model:
## Shannon ~ sex + month + (1 | mousegroup.x) + (1 | mouse)
##         Sum Sq Mean Sq NumDF  DenDF F value   Pr(>F)
## sex    0.54175 0.54175     1  87.59  9.9032 0.002255 **
## month  1.70280 0.12163    14 451.47  2.2234 0.006513 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
kable(anova(model1, type = 3), caption = "ANOVA for model1 with all cohort")
```

Table 1: ANOVA for model1 with all cohort

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.5417463	0.5417463	1	87.58639	9.903183	0.0022552
month	1.7027978	0.1216284	14	451.47031	2.223381	0.0065132

```r
kable(summary(model1)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.9071	0.0373	15.0432	104.7698	0.0000
sexM	0.0651	0.0207	87.5864	3.1469	0.0023
monthM05	-0.0496	0.0475	542.2897	-1.0436	0.2971
monthM06	0.0029	0.0522	510.3183	0.0558	0.9555
monthM07	0.0345	0.0458	533.0349	0.7531	0.4517
monthM09	-0.0184	0.0392	559.6266	-0.4684	0.6397
monthM11	0.0512	0.0464	537.2349	1.1036	0.2703
monthM12	0.0101	0.0533	515.3755	0.1898	0.8496
monthM13	0.1004	0.0491	549.2311	2.0428	0.0415
monthM15	0.0361	0.0394	553.2882	0.9142	0.3610
monthM17	0.0164	0.0502	555.3665	0.3260	0.7446
monthM18	-0.0342	0.0570	532.3179	-0.6007	0.5483
monthM19	0.0724	0.0497	551.2145	1.4564	0.1459
monthM21	-0.0680	0.0487	552.5035	-1.3971	0.1630
monthM22	-0.1553	0.0595	554.2110	-2.6079	0.0094
monthM23	-0.0694	0.0457	537.2401	-1.5191	0.1293

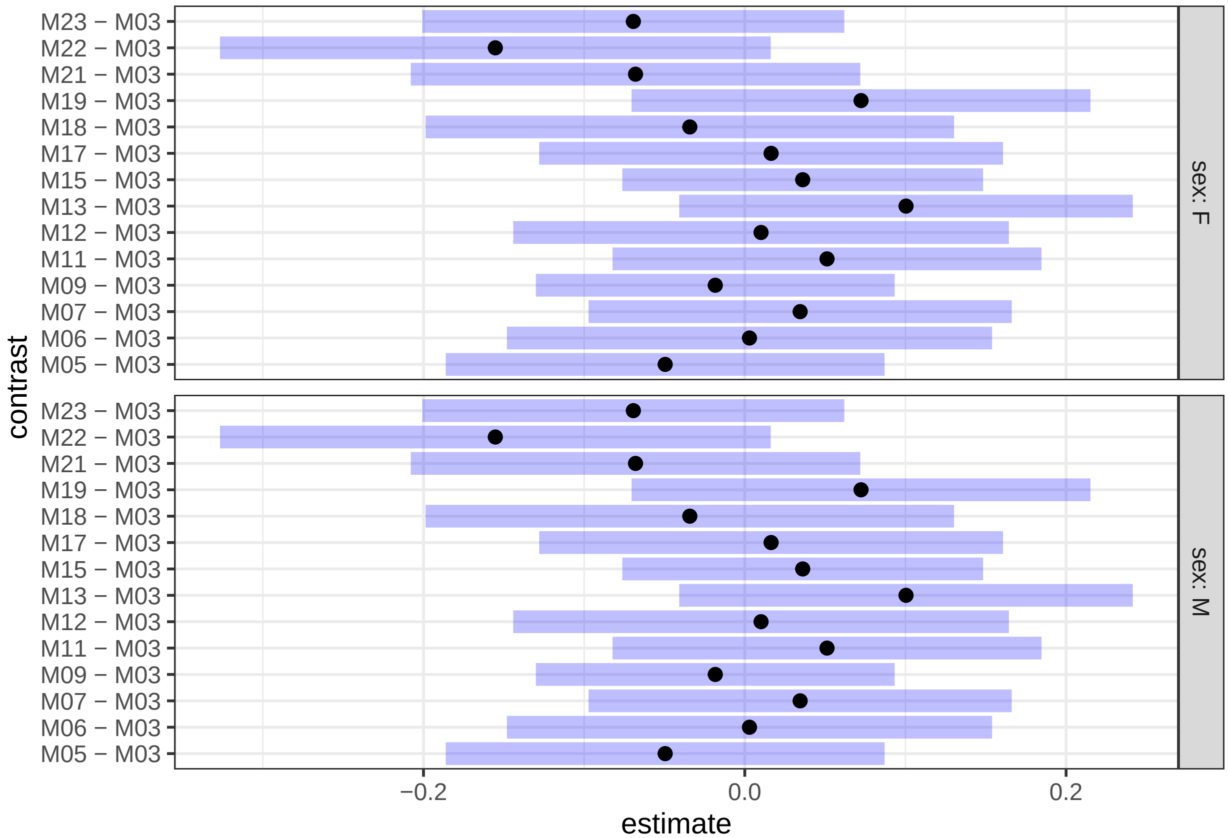
Estimated marginal means

I did this to double check the above results. And we see similar findings.

```r
# Estimated marginal means by Sex within Month for Shannon
## Q: Within each sex, how does each month differ from the reference month?

emm_month_within_sex <- emmeans(model1, ~ month | sex)
baseline_contrasts <- contrast(emm_month_within_sex, method = "trt.vs.ctrl", ref = 1, by = "sex")
baseline_contrasts
```

```
## sex = F:
##  contrast  estimate     SE  df t.ratio p.value
##  M05 - M03 -0.04956 0.0480 538  -1.033  0.9039
##  M06 - M03  0.00291 0.0531 507   0.055  1.0000
##  M07 - M03  0.03446 0.0463 529   0.745  0.9742
##  M09 - M03 -0.01837 0.0392 555  -0.468  0.9967
##  M11 - M03  0.05119 0.0469 533   1.091  0.8819
##  M12 - M03  0.01012 0.0542 512   0.187  1.0000
##  M13 - M03  0.10037 0.0496 545   2.023  0.3275
##  M15 - M03  0.03605 0.0395 548   0.914  0.9402
##  M17 - M03  0.01637 0.0507 552   0.323  0.9994
##  M18 - M03 -0.03422 0.0578 529  -0.592  0.9905
##  M19 - M03  0.07236 0.0502 548   1.442  0.6998
##  M21 - M03 -0.06802 0.0492 549  -1.384  0.7353
##  M22 - M03 -0.15529 0.0602 551  -2.578  0.1014
##  M23 - M03 -0.06937 0.0461 533  -1.503  0.6608
##
## sex = M:
##  contrast  estimate     SE  df t.ratio p.value
##  M05 - M03 -0.04956 0.0480 538  -1.033  0.9039
##  M06 - M03  0.00291 0.0531 507   0.055  1.0000
##  M07 - M03  0.03446 0.0463 529   0.745  0.9742
##  M09 - M03 -0.01837 0.0392 555  -0.468  0.9967
##  M11 - M03  0.05119 0.0469 533   1.091  0.8819
##  M12 - M03  0.01012 0.0542 512   0.187  1.0000
##  M13 - M03  0.10037 0.0496 545   2.023  0.3275
##  M15 - M03  0.03605 0.0395 548   0.914  0.9402
##  M17 - M03  0.01637 0.0507 552   0.323  0.9994
##  M18 - M03 -0.03422 0.0578 529  -0.592  0.9905
##  M19 - M03  0.07236 0.0502 548   1.442  0.6998
##  M21 - M03 -0.06802 0.0492 549  -1.384  0.7353
##  M22 - M03 -0.15529 0.0602 551  -2.578  0.1014
##  M23 - M03 -0.06937 0.0461 533  -1.503  0.6608
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: dunnettx method for 14 tests
```

```r
plot(baseline_contrasts)
```



Observed index model

```
model2_interaction <- lmer(Observed ~ sex * month + (1 | mousegroup.x) + (1 | mouse), data = df_wide)
drop1(model2_interaction,test="Chisq")
```

```
## Single term deletions using Satterthwaite's method:
##
## Model:
## Observed ~ sex * month + (1 | mousegroup.x) + (1 | mouse)
##            Sum Sq Mean Sq NumDF  DenDF F value Pr(>F)
## sex:month 7829.7  559.26    14 566.93  1.1434 0.3163
```

ANOVA for model1 (Observed with all cohorts)

```
model2 <- lmer(Observed ~ sex + month + (1 | mousegroup.x) + (1 | mouse), data = df_wide)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
drop1(model2,test="Chisq")
```

```
## Single term deletions using Satterthwaite's method:
##
## Model:
## Observed ~ sex + month + (1 | mousegroup.x) + (1 | mouse)
##         Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## sex      838.2   838.2     1 605.55  1.7045    0.1922
## month 30364.0  2168.9    14 576.51  4.4104 1.58e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
kable(anova(model2, type = 3), caption = "ANOVA for model1 with all cohort")
```

Table 3: ANOVA for model1 with all cohort

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	838.1995	838.1995	1	605.5486	1.704502	0.1921961
month	30364.0053	2168.8575	14	576.5123	4.410431	0.0000002

```r
kable(summary(model2)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	136.7254	4.3175	7.5525	31.6679	0.0000
sexM	2.3485	1.7988	605.5486	1.3056	0.1922
monthM05	10.1818	4.5122	606.1792	2.2565	0.0244
monthM06	6.5411	4.9747	600.6554	1.3149	0.1891
monthM07	22.3651	4.3518	605.5256	5.1392	0.0000
monthM09	8.7212	3.7072	604.1640	2.3525	0.0190
monthM11	13.9406	4.4105	605.7887	3.1608	0.0017
monthM12	7.5762	5.0827	601.3926	1.4906	0.1366
monthM13	23.1150	4.6646	606.5663	4.9554	0.0000
monthM15	11.0327	3.7312	604.2336	2.9569	0.0032
monthM17	7.1277	4.7653	606.5758	1.4958	0.1352
monthM18	10.3435	5.4190	603.4014	1.9087	0.0568
monthM19	14.7282	4.7164	606.4719	3.1228	0.0019
monthM21	1.9692	4.6208	606.5365	0.4262	0.6701
monthM22	-9.3238	5.6502	605.8463	-1.6502	0.0994
monthM23	6.4434	4.3417	606.0904	1.4841	0.1383

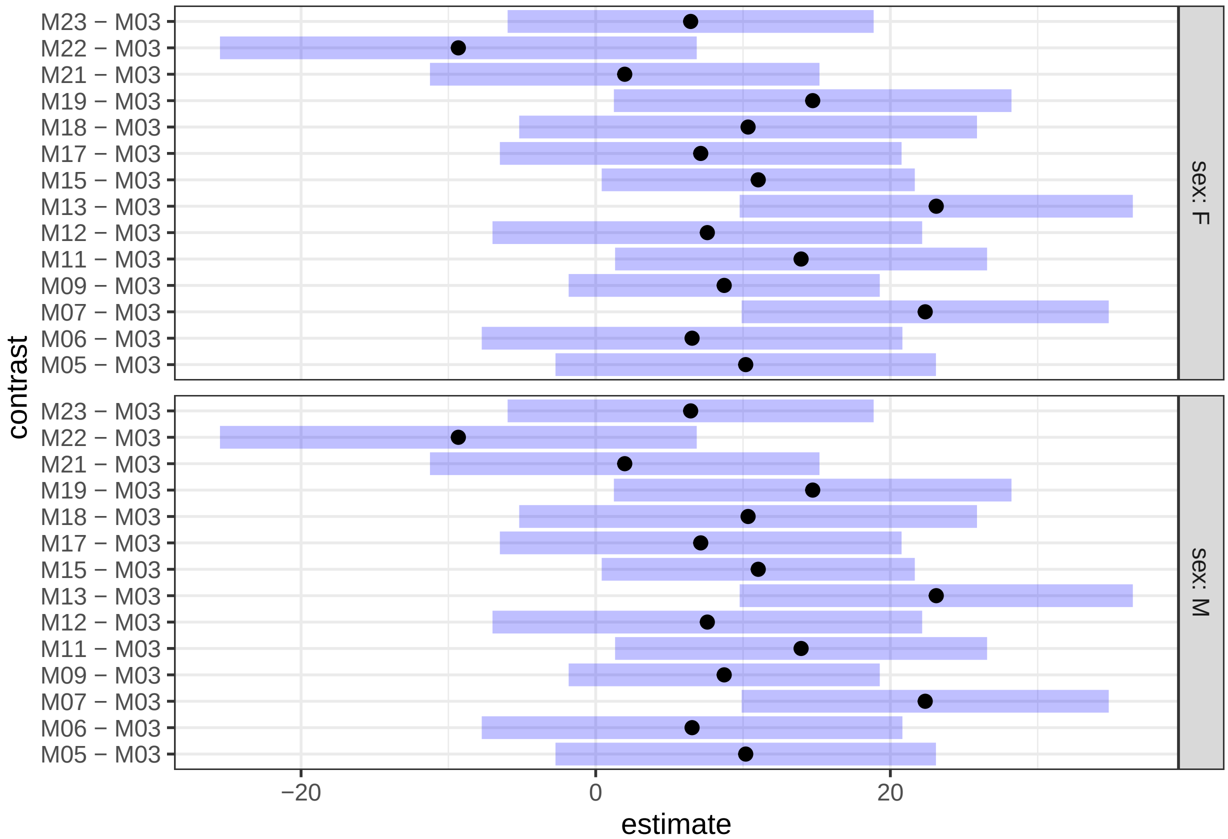
Estimated marginal means for model2 (Observed)

I did this to double check the above results. And we see similar findings.

```r
# Estimated marginal means by Sex within Month for Observed
## Q: Within each sex, how does each month differ from the reference month?

emm_month_within_sex_observed <- emmeans(model2, ~ month | sex)
baseline_contrasts_observed <- contrast(emm_month_within_sex_observed, method = "trt.vs.ctrl", ref = 1,
baseline_contrasts_observed
```

```
## sex = F:
##  contrast  estimate   SE  df t.ratio p.value
##  M05 - M03    10.18 4.54 553   2.244  0.2159
##  M06 - M03     6.54 5.02 542   1.304  0.7807
##  M07 - M03    22.37 4.38 546   5.109  <.0001
##  M09 - M03     8.72 3.71 559   2.351  0.1724
##  M11 - M03    13.94 4.44 549   3.142  0.0207
##  M12 - M03     7.58 5.12 545   1.479  0.6767
##  M13 - M03    23.11 4.69 558   4.930  <.0001
##  M15 - M03    11.03 3.73 551   2.955  0.0366
##  M17 - M03     7.13 4.79 565   1.488  0.6709
##  M18 - M03    10.34 5.46 556   1.895  0.4041
##  M19 - M03    14.73 4.74 561   3.106  0.0232
##  M21 - M03     1.97 4.65 562   0.424  0.9979
##  M22 - M03    -9.32 5.69 570  -1.640  0.5711
##  M23 - M03     6.44 4.37 546   1.476  0.6783
##
## sex = M:
##  contrast  estimate   SE  df t.ratio p.value
##  M05 - M03    10.18 4.54 553   2.244  0.2159
##  M06 - M03     6.54 5.02 542   1.304  0.7807
##  M07 - M03    22.37 4.38 546   5.109  <.0001
##  M09 - M03     8.72 3.71 559   2.351  0.1724
##  M11 - M03    13.94 4.44 549   3.142  0.0207
##  M12 - M03     7.58 5.12 545   1.479  0.6767
##  M13 - M03    23.11 4.69 558   4.930  <.0001
##  M15 - M03    11.03 3.73 551   2.955  0.0366
##  M17 - M03     7.13 4.79 565   1.488  0.6709
##  M18 - M03    10.34 5.46 556   1.895  0.4041
##  M19 - M03    14.73 4.74 561   3.106  0.0232
##  M21 - M03     1.97 4.65 562   0.424  0.9979
##  M22 - M03    -9.32 5.69 570  -1.640  0.5711
##  M23 - M03     6.44 4.37 546   1.476  0.6783
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: dunnettx method for 14 tests
```

```r
plot(baseline_contrasts_observed)
```



Alpha Diversity of mice based on frailty

To determine potential relationship between frailty scores and survival, I filtered the dataset to the last time point of collection

```
metadata<-read.delim("Young_aging_metadata_v4.txt")

# keeping only the last one per mouse
metadata_sub <- metadata %>%
  group_by(mouse) %>%
  arrange(month_n, .by_group = TRUE) %>%  # ensure chronological order
  slice_tail(n = 1) %>%
  ungroup()
```

This results in 186 mice

```
#Remove samples with no frailty information
metadata_sub_f<-metadata_sub %>%
  filter(frailty_physical %in% c("PF","NF","F"))

write.table(metadata_sub_f,"metadata_frailty.txt",sep="\t", quote = FALSE,
            row.names = FALSE)

qiime_frail<-qza_to_phyloseq(features="Aging-table-dada2.qza",
```

```r
                        tree="rooted-tree-merged.qza",
                        metadata="metadata_frailty.txt", taxonomy = "Aging-taxonomy-weighted_silva138-1.

sample_reads <- sample_sums(qiime_frail)
sorted_reads <- sort(sample_reads)
print(sorted_reads)
```

```
## M2201N3M22 F2198N4M22 F2202N1M22 F2202N2M22 M2226N2M17 M2214N3M21 F2199N2M19
##       2235       3122       4328       4359       4762       4831       5191
## F2198N1M22 M2200N3M22 F2198N3M22 M2201N1M22 F2199N3M22 M2201N2M22 M2200N4M22
##       5401       6219       6418       6638       7040       7872       8958
## M2197N2M22 M2212N3M13 M2197N1M22 M2215N2M21 F2211N3M23 F2224N2M22 M2226N3M17
##       8997      10442      10554      10873      13653      16321      16495
## F2223N1M22 F2213N3M23 M2215N1M23 M2214N2M23 M2225N1M22 M2226N4M22 M2226N1M22
##      17012      17486      17781      17844      18262      18311      19661
## M2225N3M22 M2214N1M23 M2225N2M23 F2223N2M22 M2215N3M23 M2225N5M22 F2211N4M23
##      20406      21549      22186      22423      22656      25894      27221
## F2213N1M23 F2224N3M22 M2225N4M22 F2211N1M23 M2212N1M23 F2224N4M22
##      29614      31255      34634      35112      37080      43680
```

```r
save(qiime_frail, file="qiime_frail.RData")

qiime.rare.frail<- rarefy_even_depth(qiime_frail,sample.size = min(sample_sums(qiime_frail)))
```

```
## You set `rngseed` to FALSE. Make sure you've set & recorded
##  the random seed of your session for reproducibility.
## See `?set.seed`

## ...

## 3896OTUs were removed because they are no longer
## present in any sample after random subsampling

## ...
```

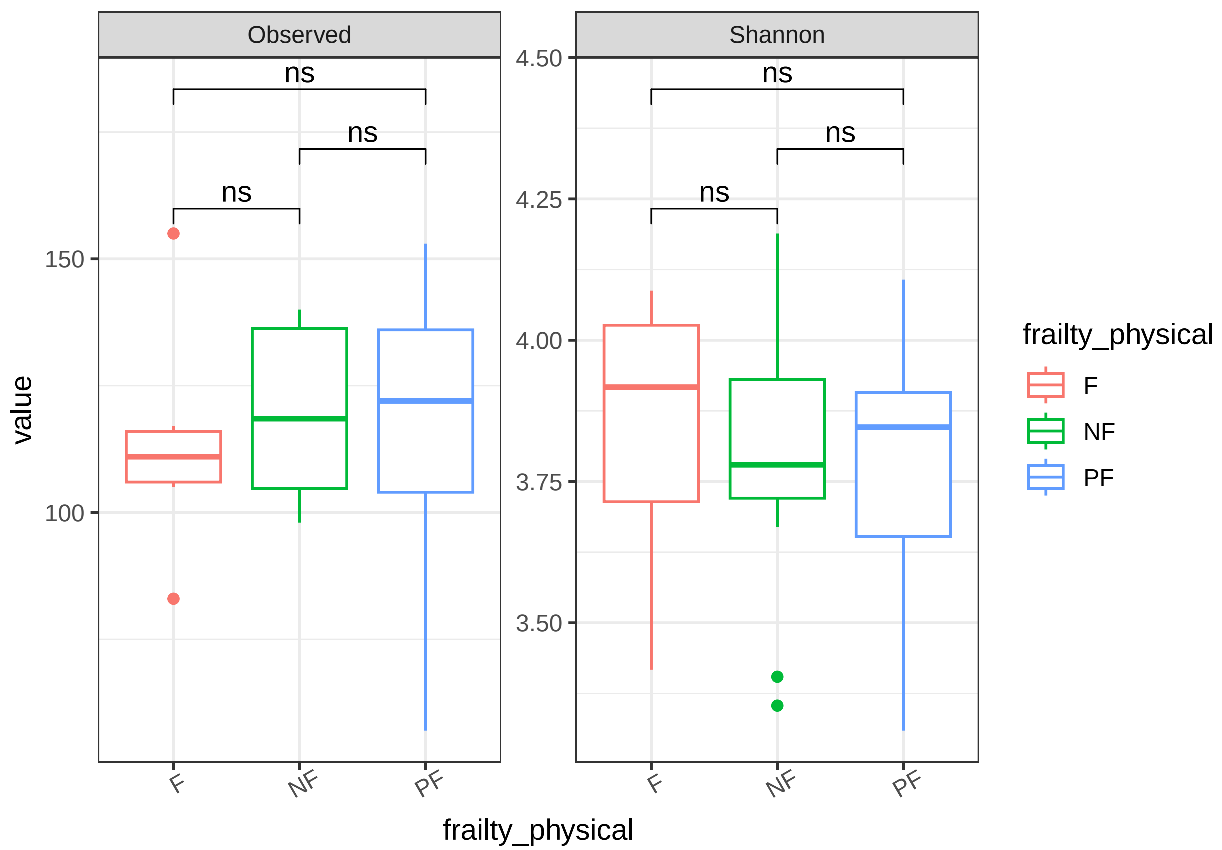
Alpha diversity

```r
metadata_df_frail<-as.data.frame(sample_data( qiime.rare.frail))
metadata_df_frail$SampleID<-rownames(metadata_df_frail)
metadata_df_frail<-as.matrix(metadata_df_frail)

alpha.div.df.frail <- estimate_richness(
  qiime.rare.frail,
  measures=c("Shannon", "Observed")) %>%
  rownames_to_column("SampleID") %>%
  merge(metadata_df_frail, by="SampleID")

alpha.div.df.frail.long <- alpha.div.df.frail %>%
```

```r
  pivot_longer(c(Shannon, Observed), names_to="metric")

alpha.div.plot.df.frail <- alpha.div.df.frail.long %>%
  arrange(metric, frailty_physical)

a_my_comparisons_frail <-list(c("F","NF"),c("NF","PF"),c("PF","F"))
symnum.args <- list(cutpoints = c(0, 0.0001, 0.001, 0.01, 0.05, 1), symbols = c("****", "***", "**", "*

alpha.div.df.frail.long%>%
  ggplot(aes(x=frailty_physical, y=value, color=frailty_physical))+
geom_boxplot()+facet_wrap(~metric, scales="free")+
  theme_set(theme_bw())+
  theme(axis.text.x=element_text(angle=30))+
  stat_compare_means(comparisons = a_my_comparisons_frail, symnum.args = symnum.args, method = "wilcox.
```

I found no association with frailty scores and microbiota

```
## Warning in wilcox.test.default(c(105, 117, 83, 113, 155, 109), c(101, 136, :
## cannot compute exact p-value with ties

## Warning in wilcox.test.default(c(101, 136, 107, 140, 98, 106, 138, 124, :
## cannot compute exact p-value with ties

## Warning in wilcox.test.default(c(107, 82, 92, 84, 101, 139, 153, 135, 121, :
## cannot compute exact p-value with ties
```



```r
#No. sig difference found here
```

```r
df_wide_frail<- alpha.div.df.frail.long%>%
  pivot_wider(names_from = metric, values_from = value)
model_shannon_frail <- lmer(Shannon ~ frailty_physical + (1 | mousegroup.x), data = df_wide_frail)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
kable(summary(model_shannon_frail)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8414	0.0919	38	41.7782	0.0000
frailty_physicalNF	-0.0559	0.1126	38	-0.4965	0.6224
frailty_physicalPF	-0.0655	0.1032	38	-0.6344	0.5296

```r
df_wide_frail$howlette_frailty<-as.numeric(df_wide_frail$howlette_frailty)
model_shannon_hf <- lmer(Shannon ~ howlette_frailty + (1 | mousegroup.x), data = df_wide_frail)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
kable(summary(model_shannon_hf)$coefficients, digits = 4)
```

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8460	0.1026	39	37.4759	0.0000
howlette_frailty	-0.2416	0.4041	39	-0.5980	0.5533

Subsetting per cohort (just a sanity check)

Cohort G1

```r
df_wide_G1<- df_wide %>%
  filter(mousegroup.x== "G1")
View(df_wide_G1)

model1_G1 <- lmer(Shannon ~ sex + month + (1 | mouse), data = df_wide_G1)
summary(model1_G1)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Shannon ~ sex + month + (1 | mouse)
##    Data: df_wide_G1
##
## REML criterion at convergence: 11.2
##
## Scaled residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -2.7847 -0.6113  0.2098  0.6180  2.2660
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  mouse    (Intercept) 0.00000  0.0000
##  Residual             0.05076  0.2253
## Number of obs: 142, groups:  mouse, 19
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)   3.939675   0.060277 131.000000  65.359   <2e-16 ***
## sexM          0.038719   0.038174 131.000000   1.014    0.312
## monthM05      0.020445   0.088337 131.000000   0.231    0.817
## monthM07     -0.046386   0.079653 131.000000  -0.582    0.561
## monthM09     -0.021581   0.079688 131.000000  -0.271    0.787
## monthM11     -0.029638   0.078551 131.000000  -0.377    0.707
## monthM13      0.134364   0.084127 131.000000   1.597    0.113
## monthM15      0.064583   0.081052 131.000000   0.797    0.427
## monthM17      0.098800   0.081206 131.000000   1.217    0.226
## monthM19      0.107428   0.082449 131.000000   1.303    0.195
## monthM22     -0.004881   0.093956 131.000000  -0.052    0.959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) sexM   mntM05 mntM07 mntM09 mntM11 mntM13 mntM15 mntM17 mntM19
## sexM     -0.356
## monthM05 -0.612  0.047
## monthM07 -0.661  0.000  0.451
## monthM09 -0.671  0.030  0.452  0.500
## monthM11 -0.686  0.045  0.459  0.507  0.508
## monthM13 -0.629  0.011  0.427  0.473  0.474  0.481
## monthM15 -0.665  0.045  0.445  0.491  0.492  0.500  0.466
## monthM17 -0.675  0.076  0.446  0.490  0.492  0.501  0.465  0.485
## monthM19 -0.637 -0.004  0.435  0.483  0.483  0.490  0.457  0.475  0.473
## monthM22 -0.545 -0.042  0.380  0.424  0.422  0.428  0.401  0.415  0.413  0.410
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
drop1(model1_G1,test="Chisq")
```

```
## Single term deletions using Satterthwaite's method:
##
## Model:
## Shannon ~ sex + month + (1 | mouse)
##        Sum Sq  Mean Sq NumDF DenDF F value Pr(>F)
## sex   0.05222 0.052216     1   131  1.0288 0.3123
## month 0.53683 0.059648     9   131  1.1752 0.3161
```

```r
# From the summary result: There was no difference between sexes at M03, or difference with females, or
```

```r
kable(anova(model1_G1, type = 3), caption ="ANOVA for Cohort G1: Shannon")
```

Table 7: ANOVA for Cohort G1: Shannon

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.0522157	0.0522157	1	131	1.028756	0.3123207
month	0.5368291	0.0596477	9	131	1.175181	0.3160914

```r
model2_G1 <- lmer(Observed ~ sex * month + (1 | mouse), data = df_wide_G1)
```

```
## boundary (singular) fit: see help('isSingular')
```

cohort G1: Observed

From the below table:

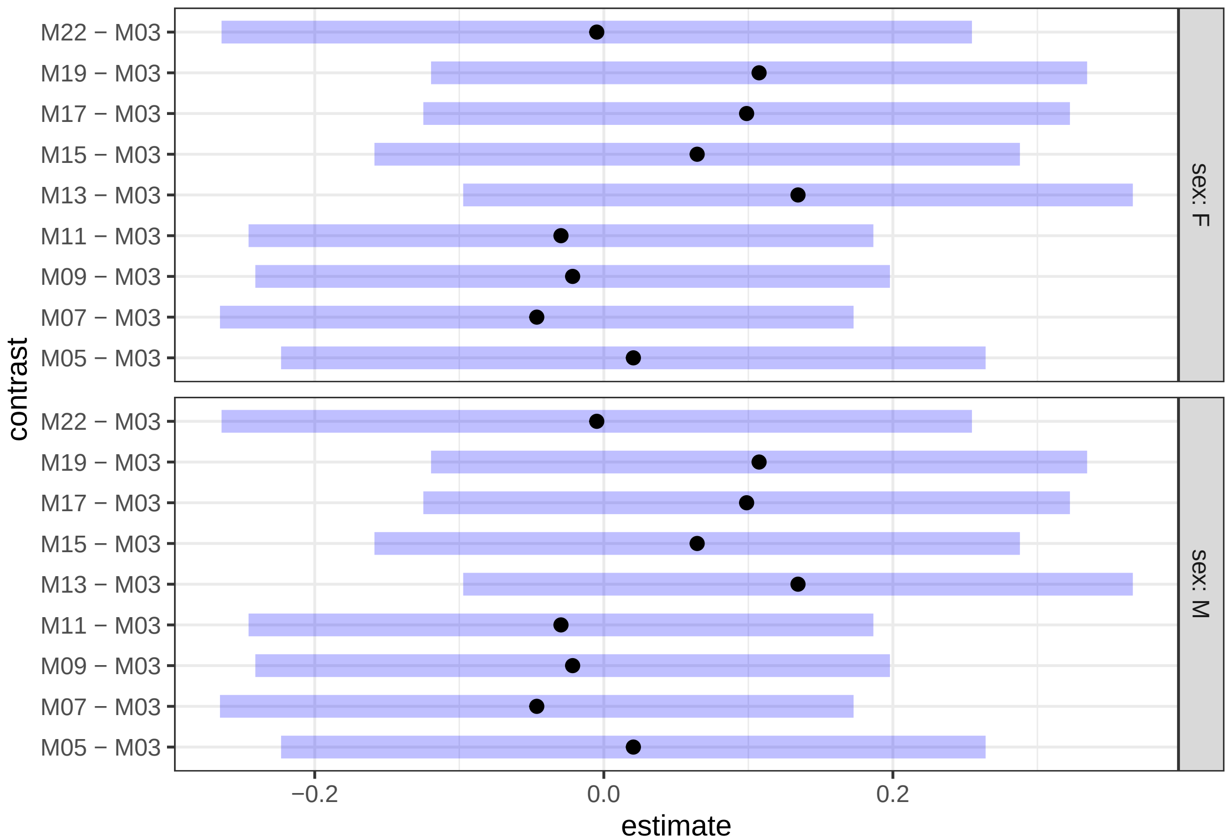
1. There is no difference between male and females at M03
2. Females at month M07, M09, M13, M16, and M17 are significantly different from females at M03
3. Compared to the difference between males and females in M03, there was no sig difference between Males and Females in other months

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	123.1429	8.4062	122	14.6491	0.0000
sexM	2.3016	11.2082	122	0.2053	0.8376
monthM05	5.5238	12.3736	122	0.4464	0.6561
monthM07	45.4286	11.8881	122	3.8213	0.0002
monthM09	45.8571	11.5106	122	3.9839	0.0001
monthM11	16.9683	11.2082	122	1.5139	0.1326
monthM13	46.1905	12.3736	122	3.7330	0.0003
monthM15	32.1071	11.5106	122	2.7893	0.0061
monthM17	28.6349	11.2082	122	2.5548	0.0119
monthM19	7.0238	12.3736	122	0.5676	0.5713
monthM22	-14.4762	15.3475	122	-0.9432	0.3474
sexM:monthM05	-8.7683	17.5213	122	-0.5004	0.6177
sexM:monthM07	-6.4286	15.8509	122	-0.4056	0.6858
sexM:monthM09	-15.0516	15.7888	122	-0.9533	0.3423
sexM:monthM11	8.2123	15.5697	122	0.5275	0.5988
sexM:monthM13	-2.4921	16.6952	122	-0.1493	0.8816
sexM:monthM15	-12.5516	16.0661	122	-0.7812	0.4362
sexM:monthM17	-15.0794	16.2181	122	-0.9298	0.3543
sexM:monthM19	28.5317	16.4285	122	1.7367	0.0850
sexM:monthM22	17.1984	19.3119	122	0.8906	0.3749

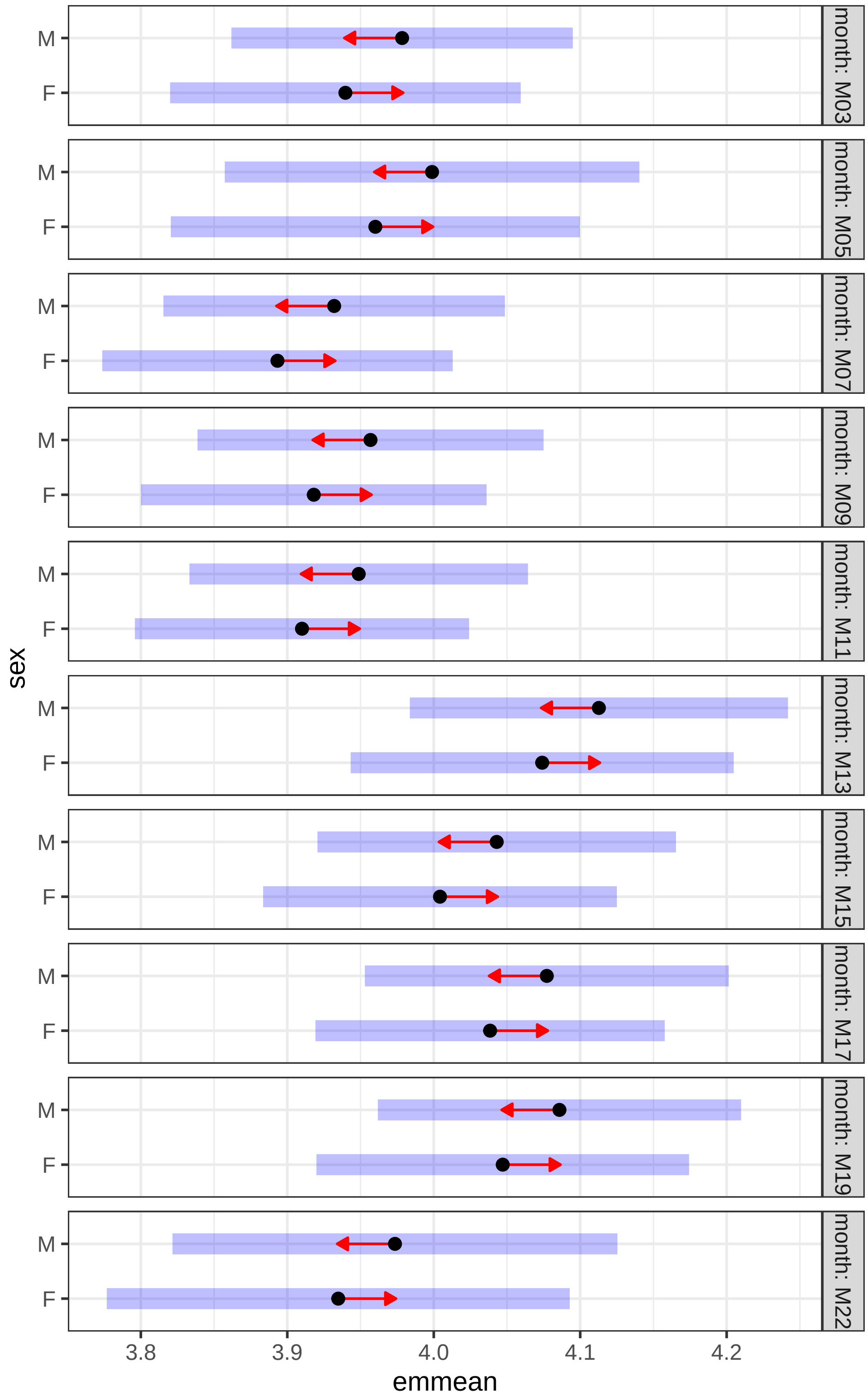
ANOVA of ModelG1: Observed

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	91.47856	91.47856	1	122	0.1849369	0.6679231
month	36277.26342	4030.80705	9	122	8.1488478	2.09e-09 ***
sex:month	6297.16741	699.68527	9	122	1.4145130	0.1890481

```
# Estimated marginal means by Sex within Month for Shannon
emm_sex_within_month_G1 <- emmeans(model1_G1, ~ sex | month)
pairs_sex_within_month_G1 <- contrast(emm_sex_within_month_G1, method = "pairwise", adjust = "holm")

emm_month_within_sex_G1 <- emmeans(model1_G1, ~ month | sex)
baseline_contrasts_G1 <- contrast(emm_month_within_sex_G1, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_G1)
```




```
plot(emm_sex_within_month_G1, comparisons = TRUE)
```



M
F
month: M03
month: M05
month: M07
month: M09
month: M11
month: M13
month: M15
month: M17
month: M19
month: M22
sex
3.8
3.9
4.0
4.1
4.2
emmean

Cohort C

```r
df_wide_C<- df_wide %>%
  filter(mousegroup.x== "C")
model1_C <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_C)
```

Summary Cohort C: Shannon

From the below table:

1. There is sig. difference between male and females at M03 (p=0.0005)
2. Females shannon diversity relatively stable with age
3. Compared to the difference between males and females in M03, there was sig difference (decreasing) between Males and Females in months M12 (p= 0.0347), M15 (p=0.0320), and M23 (p=0.0006)

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.8023	0.0608	219.0609	62.5293	0.0000
sexM	0.2902	0.0821	218.7018	3.5329	0.0005
monthM06	0.0473	0.0810	193.8009	0.5838	0.5600
monthM09	0.0858	0.0792	201.6122	1.0829	0.2802
monthM12	0.1241	0.0838	196.9755	1.4807	0.1403
monthM15	0.0797	0.0810	193.8009	0.9837	0.3265
monthM18	0.0662	0.0891	193.1763	0.7423	0.4588
monthM21	0.0266	0.0822	192.6006	0.3238	0.7464
monthM23	0.1353	0.0798	192.4630	1.6951	0.0917
sexM:monthM06	-0.1082	0.1128	195.5182	-0.9591	0.3387
sexM:monthM09	-0.1615	0.1170	202.4634	-1.3803	0.1690
sexM:monthM12	-0.2441	0.1148	196.0534	-2.1268	0.0347
sexM:monthM15	-0.2437	0.1128	195.9673	-2.1600	0.0320
sexM:monthM18	-0.2234	0.1210	196.4263	-1.8459	0.0664
sexM:monthM21	-0.2252	0.1161	198.1406	-1.9398	0.0538
sexM:monthM23	-0.3868	0.1101	193.5140	-3.5121	0.0006

```r
kable(anova(model1_C, type = 3), caption = "ANOVA for model1 (Shannon)_C")
```

Table 11: ANOVA for model1 (Shannon)_C

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
sex	0.3586323	0.3586323	1	36.60868	7.345232	0.0101709
month	0.2028897	0.0289842	7	197.69699	0.593633	0.7606966
sex:month	0.7006837	0.1000977	7	197.69699	2.050124	0.0506958

```r
model2_C <- lmer(Observed ~ sex * month + (1 | mouse), data = df_wide_C)
```

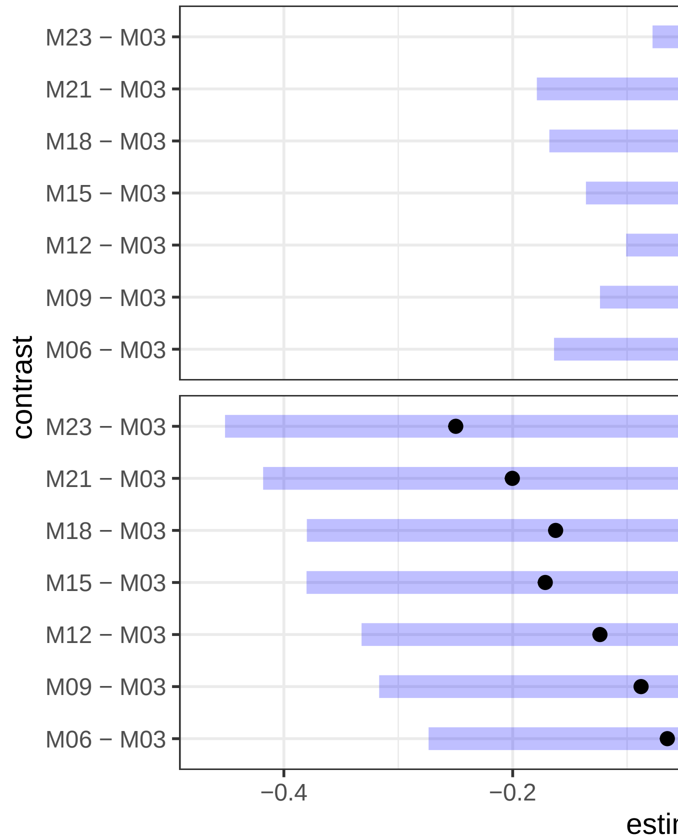
```
## boundary (singular) fit: see help('isSingular')
```

```r
kable(summary(model2_C)$coefficients, digits = 4, caption = "Fixed effects for Observed model (Cohort C
```

Table 12: Fixed effects for Observed model (Cohort C)

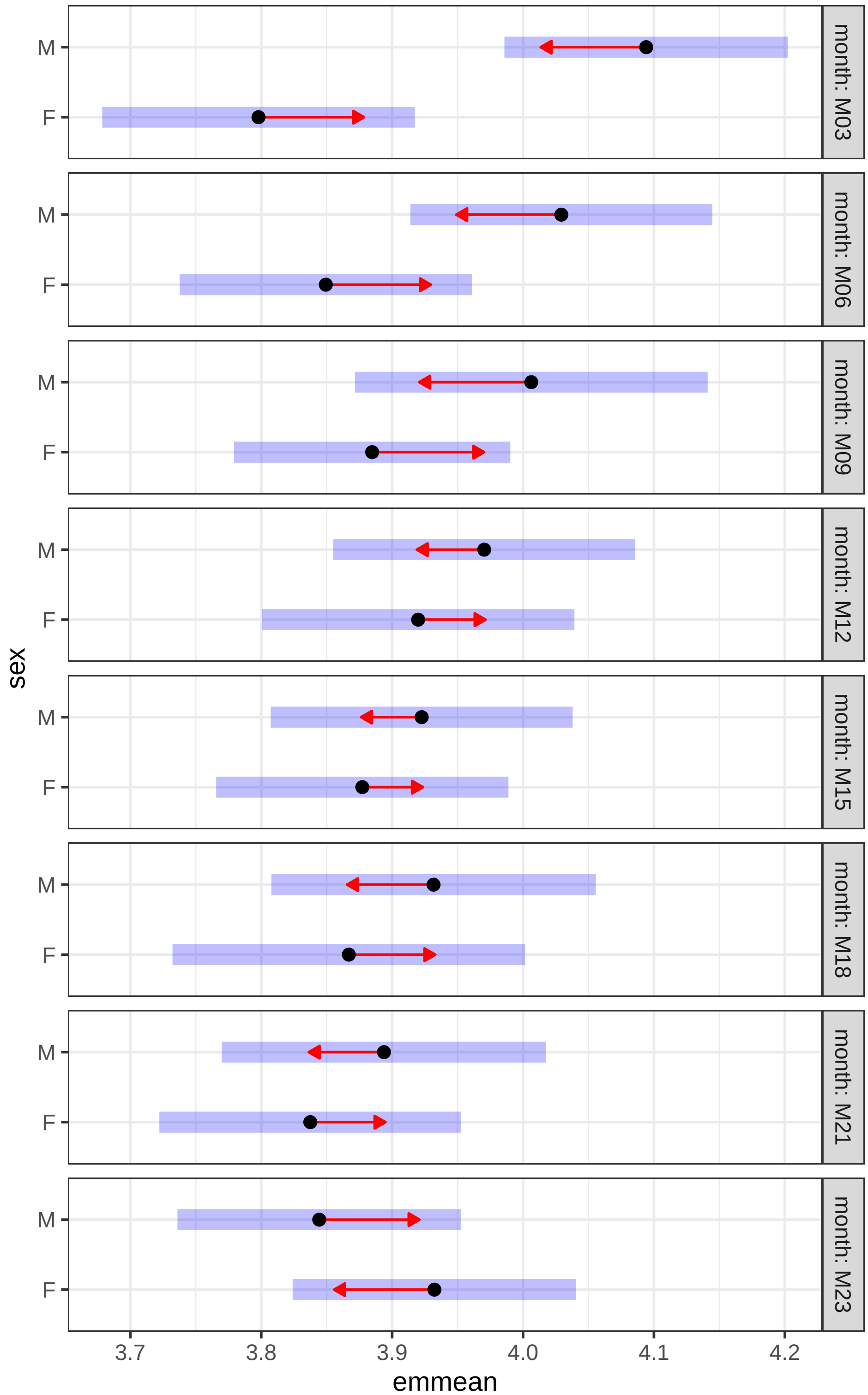
	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	140.0714	4.4457	221	31.5073	0.0000
sexM	9.4580	6.0034	221	1.5754	0.1166
monthM06	3.8661	6.0875	221	0.6351	0.5260
monthM09	1.9286	5.9276	221	0.3254	0.7452
monthM12	12.1429	6.2872	221	1.9314	0.0547
monthM15	8.9286	6.0875	221	1.4667	0.1439
monthM18	9.3831	6.7021	221	1.4000	0.1629
monthM21	9.3952	6.1815	221	1.5199	0.1300
monthM23	10.3992	6.0034	221	1.7322	0.0846
sexM:monthM06	-1.7955	8.4723	221	-0.2119	0.8324
sexM:monthM09	1.1784	8.7503	221	0.1347	0.8930
sexM:monthM12	-16.1389	8.6169	221	-1.8729	0.0624
sexM:monthM15	-11.1246	8.4723	221	-1.3131	0.1905
sexM:monthM18	-5.5279	9.0818	221	-0.6087	0.5434
sexM:monthM21	-14.1554	8.7047	221	-1.6262	0.1053
sexM:monthM23	-14.1050	8.2821	221	-1.7031	0.0900

```r
# Estimated marginal means by Sex within Month for Shannon
emm_sex_within_month_C <- emmeans(model1_C, ~ sex | month)
pairs_sex_within_month_C <- contrast(emm_sex_within_month_C, method = "pairwise", adjust = "holm")

emm_month_within_sex_C <- emmeans(model1_C, ~ month | sex)
baseline_contrasts_C <- contrast(emm_month_within_sex_C, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_C)
```



In cohort C, no differences in Observed diversity metric.

```
plot(emm_sex_within_month_C, comparisons = TRUE)
```



For Cohort G2

SHANNON

1. Sig. diff between M and F in M03
2. Females are stable
3. No differences in M and F

Table 13: Fixed effects for Observed model (Cohort G2)

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	3.7939	0.1246	101.2543	30.4592	0.0000
sexM	0.1223	0.1505	99.3598	0.8126	0.4184
monthM05	-0.2431	0.1490	92.9471	-1.6313	0.1062
monthM07	0.3279	0.1489	92.7019	2.2017	0.0302
monthM09	0.0667	0.1499	94.2028	0.4450	0.6574
monthM11	0.1733	0.1713	97.3980	1.0116	0.3142
monthM13	0.0261	0.1600	94.1260	0.1631	0.8708
monthM15	0.1091	0.1847	96.9481	0.5908	0.5560
monthM17	-0.0433	0.1544	94.8884	-0.2803	0.7799
monthM19	0.3025	0.1544	94.8884	1.9587	0.0531
monthM21	-0.0113	0.1825	94.0437	-0.0618	0.9509
monthM23	-0.0786	0.1600	94.2136	-0.4913	0.6244
sexM:monthM05	0.2209	0.1907	92.3040	1.1579	0.2499
sexM:monthM07	-0.2129	0.1995	92.9569	-1.0674	0.2886
sexM:monthM09	-0.1397	0.1915	93.3516	-0.7293	0.4677
sexM:monthM11	-0.1075	0.2087	95.8024	-0.5149	0.6078
sexM:monthM13	0.0446	0.1995	93.2802	0.2236	0.8236
sexM:monthM15	-0.1136	0.2231	95.7534	-0.5091	0.6119
sexM:monthM17	0.0029	0.2036	94.1650	0.0142	0.9887
sexM:monthM19	-0.2752	0.2108	94.4461	-1.3056	0.1949
sexM:monthM21	-0.0380	0.2256	93.7753	-0.1683	0.8667
sexM:monthM23	-0.0581	0.2079	93.9131	-0.2794	0.7805

OBSERVED For G2

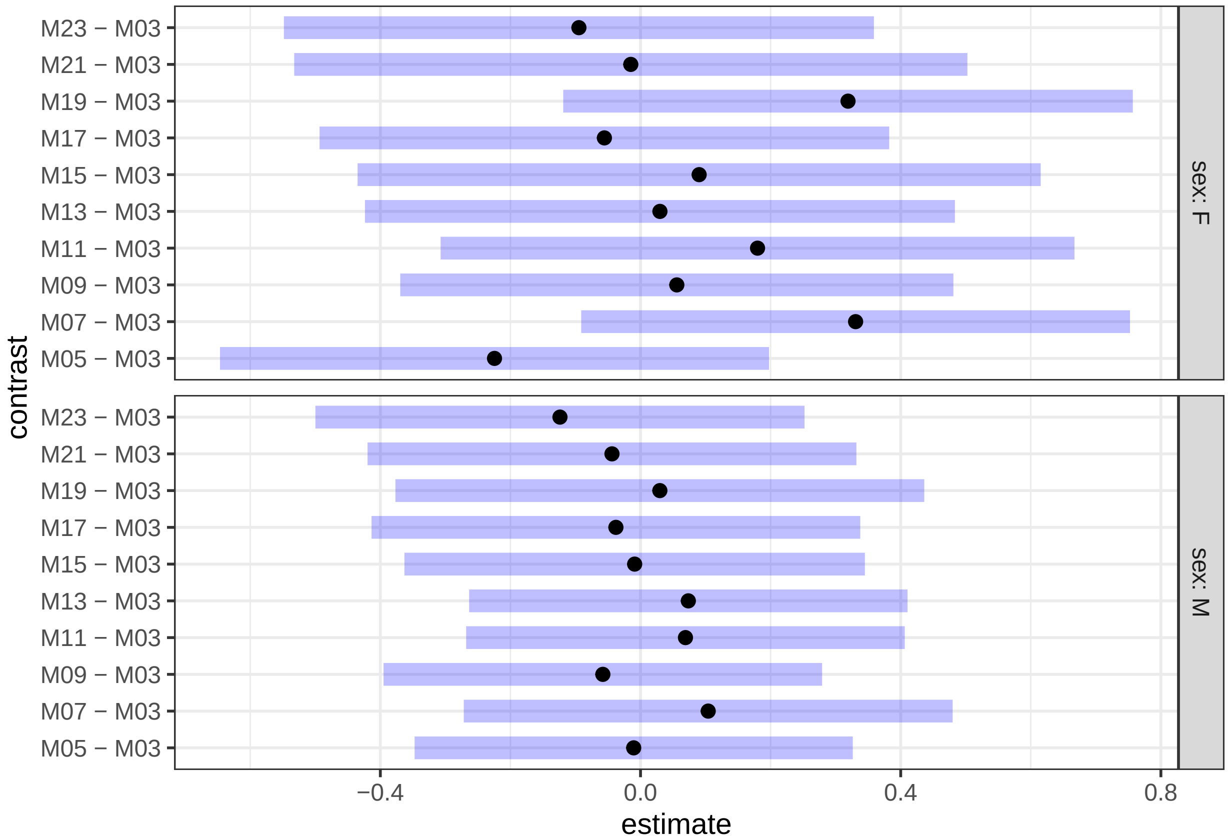
1. No diff at baseline
2. M07 and M19 females are diff from M03
3. Marginal diff between M and F in M19

Table 14: Fixed effects for Observed model (Cohort G2)

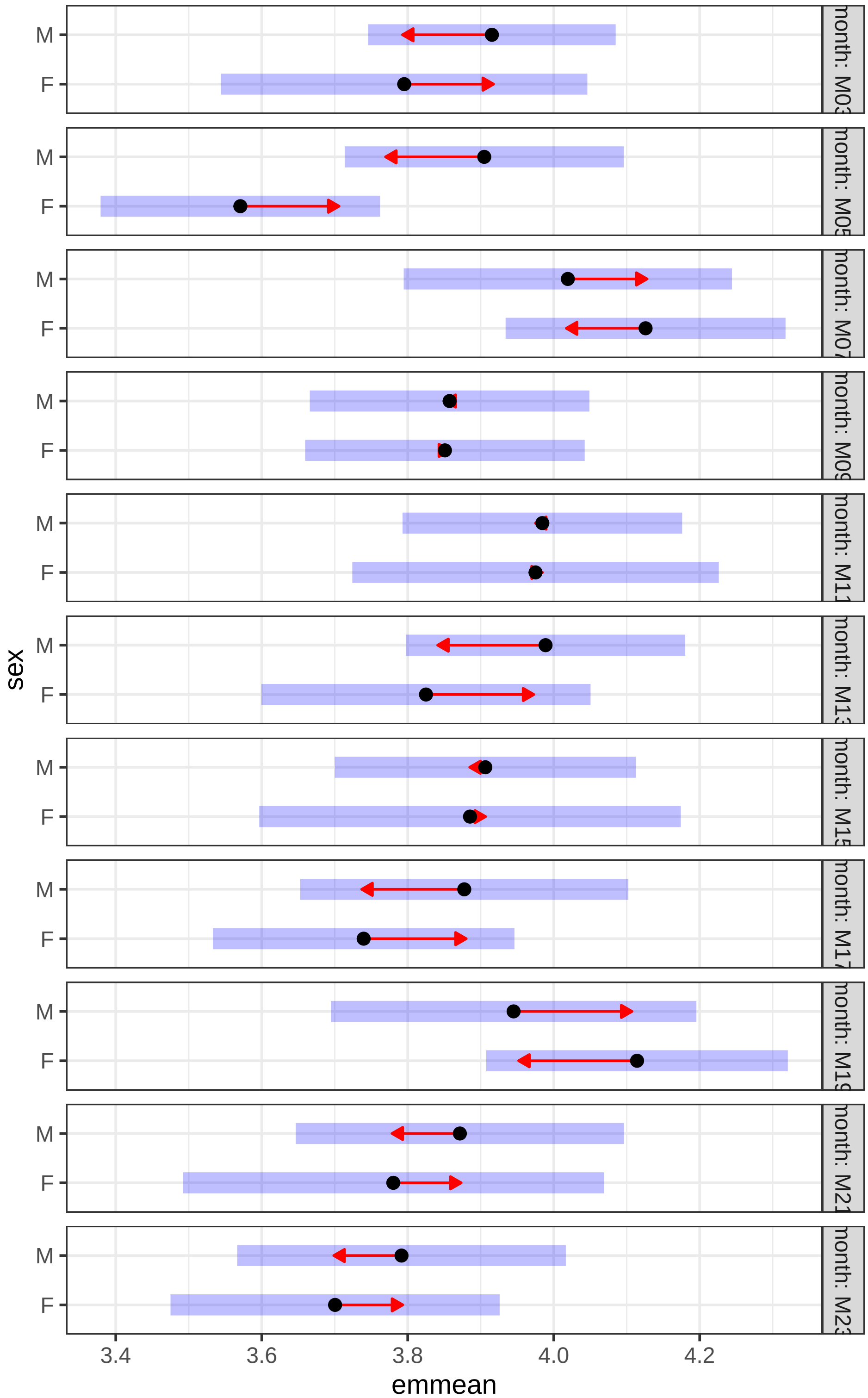
	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	116.9246	11.8680	100.9588	9.8521	0.0000
sexM	13.2977	14.3580	98.5733	0.9261	0.3566
monthM05	11.7280	14.1262	92.5827	0.8302	0.4085
monthM07	45.4176	14.1182	92.3351	3.2170	0.0018
monthM09	24.7788	14.2162	93.7734	1.7430	0.0846
monthM11	13.0188	16.2621	96.9500	0.8006	0.4253
monthM13	3.5086	15.1743	93.7729	0.2312	0.8176
monthM15	16.5746	17.5303	96.4958	0.9455	0.3468
monthM17	17.7518	14.6462	94.5244	1.2120	0.2285
monthM19	56.7518	14.6462	94.5244	3.8748	0.0002
monthM21	-10.5578	17.3059	93.6354	-0.6101	0.5433

	Estimate	Std. Error	df	t value	Pr(>|t|)
monthM23	17.0319	15.1767	93.8676	1.1222	0.2646
sexM:monthM05	15.8732	18.0796	91.9626	0.8780	0.3823
sexM:monthM07	-24.2694	18.9094	92.6070	-1.2835	0.2025
sexM:monthM09	-13.9993	18.1584	92.9841	-0.7710	0.4427
sexM:monthM11	2.7607	19.8010	95.3951	0.1394	0.8894
sexM:monthM13	15.7810	18.9142	92.9558	0.8343	0.4062
sexM:monthM15	-9.2275	21.1623	95.3389	-0.4360	0.6638
sexM:monthM17	-8.5881	19.3048	93.8117	-0.4449	0.6574
sexM:monthM19	-37.3228	19.9866	94.0765	-1.8674	0.0650
sexM:monthM21	5.2741	21.3946	93.3928	0.2465	0.8058
sexM:monthM23	-8.5674	19.7165	93.5724	-0.4345	0.6649

```
# Estimated marginal means by Sex within Month for Shannon
df_wide_G2<- df_wide %>%
  filter(mousegroup.x== "G2")

model1_G2 <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_G2)
emm_sex_within_month_G2 <- emmeans(model1_G2, ~ sex | month)
pairs_sex_within_month_G2 <- contrast(emm_sex_within_month_G2, method = "pairwise", adjust = "holm")

emm_month_within_sex_G2 <- emmeans(model1_G2, ~ month | sex)
baseline_contrasts_G2 <- contrast(emm_month_within_sex_G2, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_G2)
```




```r
plot(emm_sex_within_month_G2, comparisons = TRUE)
```



Cohort G3

OBSERVED

M11 females diff from M03 females

	Estimate	Std. Error	df	t value	Pr(>|t|)
(Intercept)	151.6000	9.3928	100	16.1401	0.0000
sexM	4.4000	11.5037	100	0.3825	0.7029
monthM05	4.6000	13.2834	100	0.3463	0.7298
monthM07	-2.2667	12.7179	100	-0.1782	0.8589
monthM09	8.4000	15.3383	100	0.5476	0.5852
monthM11	33.9000	14.0892	100	2.4061	0.0180
monthM13	11.1500	14.0892	100	0.7914	0.4306
monthM15	8.0667	15.3383	100	0.5259	0.6001
monthM17	-29.6000	23.0075	100	-1.2865	0.2012
monthM19	-5.8000	13.2834	100	-0.4366	0.6633
monthM22	-20.7667	12.7179	100	-1.6329	0.1056
monthM23	-14.0000	22.0280	100	-0.6356	0.5265
sexM:monthM05	12.5111	16.4187	100	0.7620	0.4479
sexM:monthM07	2.9667	15.8104	100	0.1876	0.8515
sexM:monthM09	-44.2571	18.5039	100	-2.3918	0.0186
sexM:monthM11	-25.0000	16.9331	100	-1.4764	0.1430
sexM:monthM13	13.3500	17.7802	100	0.7508	0.4545
sexM:monthM15	5.3778	18.1215	100	0.2968	0.7673
sexM:monthM17	24.1000	25.4357	100	0.9475	0.3457
sexM:monthM19	-13.6000	17.5723	100	-0.7739	0.4408
sexM:monthM22	13.2667	16.7146	100	0.7937	0.4292

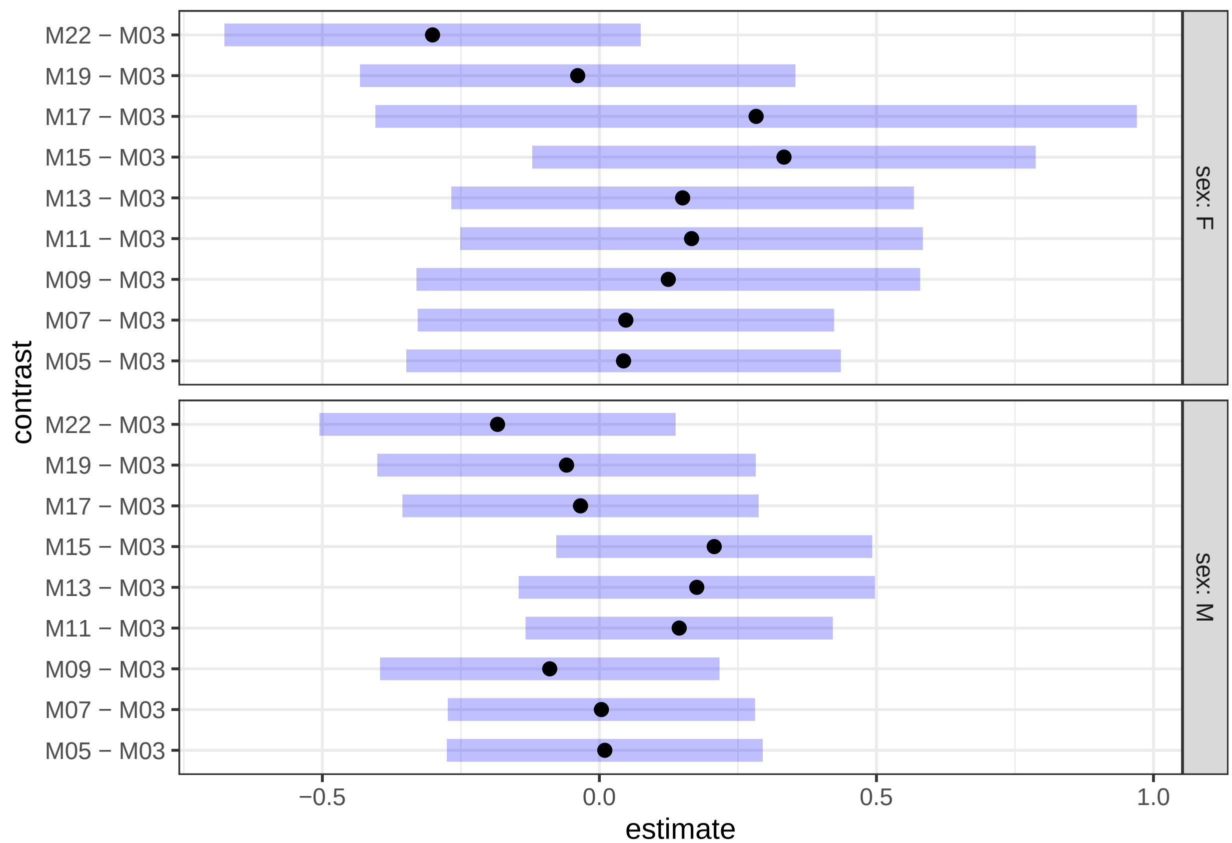
SHANNON

	Estimate	Std. Error	df	t value	Pr()
(Intercept)	3.9141	0.1000	100	39.1539	0.0000
sexM	0.0736	0.1224	100	0.6010	0.5492
monthM05	0.0475	0.1414	100	0.3358	0.7377
monthM07	0.0356	0.1354	100	0.2631	0.7930
monthM09	0.1372	0.1632	100	0.8405	0.4026
monthM11	0.1876	0.1499	100	1.2514	0.2137
monthM13	0.1749	0.1499	100	1.1666	0.2462
monthM15	0.3460	0.1632	100	2.1197	0.0365*
monthM17	0.2787	0.2449	100	1.1383	0.2577
monthM19	-0.0303	0.1414	100	-0.2143	0.8307
monthM22	-0.3130	0.1354	100	-2.3122	0.0228*
sexM:monthM05	-0.0310	0.1747	100	-0.1777	0.8594
sexM:monthM07	-0.0396	0.1683	100	-0.2354	0.8144
sexM:monthM09	-0.2277	0.1969	100	-1.1564	0.2503
sexM:monthM11	-0.0360	0.1802	100	-0.2000	0.8419
sexM:monthM13	0.0120	0.1892	100	0.0635	0.9495
sexM:monthM15	-0.1370	0.1929	100	-0.7106	0.4790
sexM:monthM17	-0.3000	0.2707	100	-1.1080	0.2705
sexM:monthM19	-0.0251	0.1870	100	-0.1342	0.8935
sexM:monthM22	0.1546	0.1779	100	0.8691	0.3869

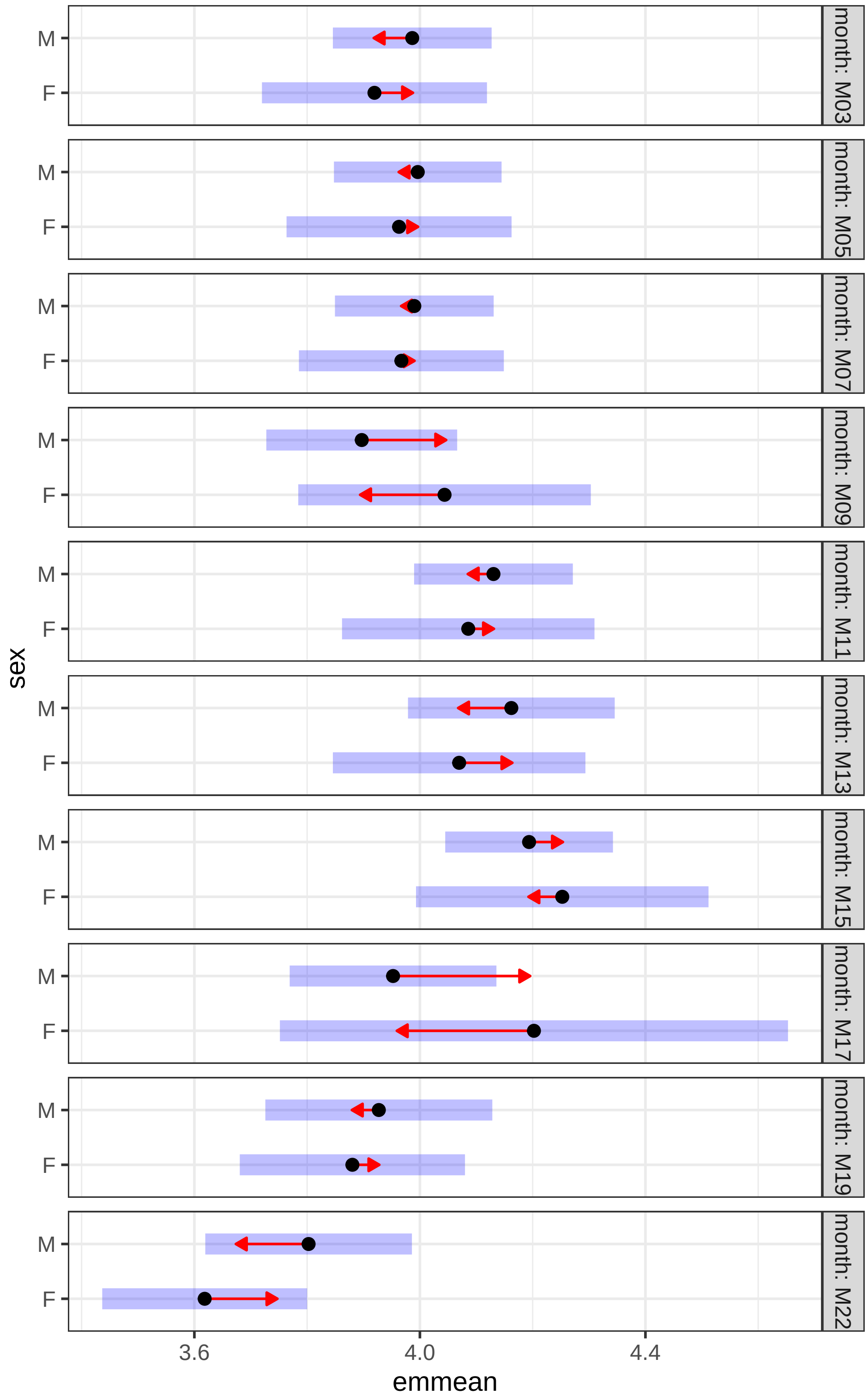
```
# Estimated marginal means by Sex within Month for Shannon
df_wide_G3<- df_wide %>%
  filter(mousegroup.x== "G3")

model1_G3 <- lmer(Shannon ~ sex * month + (1 | mouse), data = df_wide_G3)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
emm_sex_within_month_G3 <- emmeans(model1_G3, ~ sex | month)
pairs_sex_within_month_G3 <- contrast(emm_sex_within_month_G3, method = "pairwise", adjust = "holm")

emm_month_within_sex_G3 <- emmeans(model1_G3, ~ month | sex)
baseline_contrasts_G3 <- contrast(emm_month_within_sex_G3, method = "trt.vs.ctrl", ref = 1, by = "sex")

plot(baseline_contrasts_G3)
```




```r
plot(emm_sex_within_month_G3, comparisons = TRUE)
```


month: M03
month: M05
month: M07
month: M09
month: M11
month: M13
month: M15
month: M17
month: M19
month: M22
M
F
sex
3.6
4.0
4.4
emmean